



University of San Carlos | Department of
COMPUTER ENGINEERING

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P1

CpE 3104 – Microprocessors

Memory Address Decoding and Interfacing Part I

Unit 5: MEMORY INTERFACING

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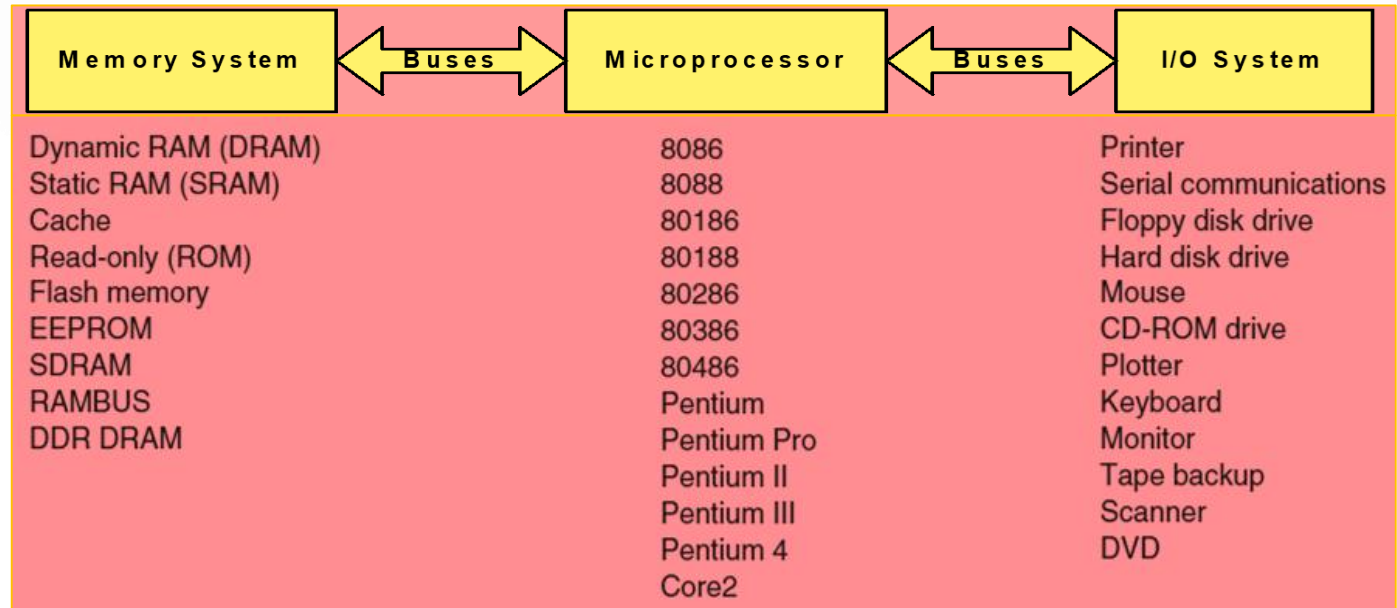
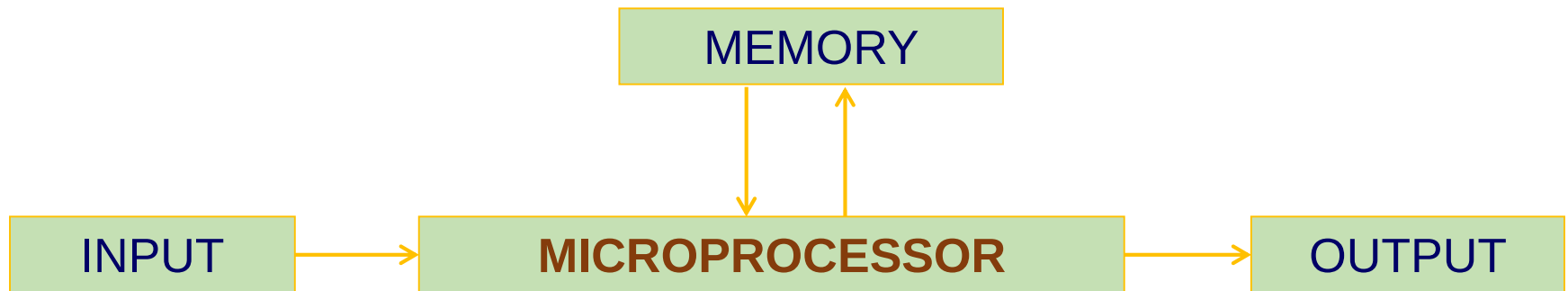
Outline



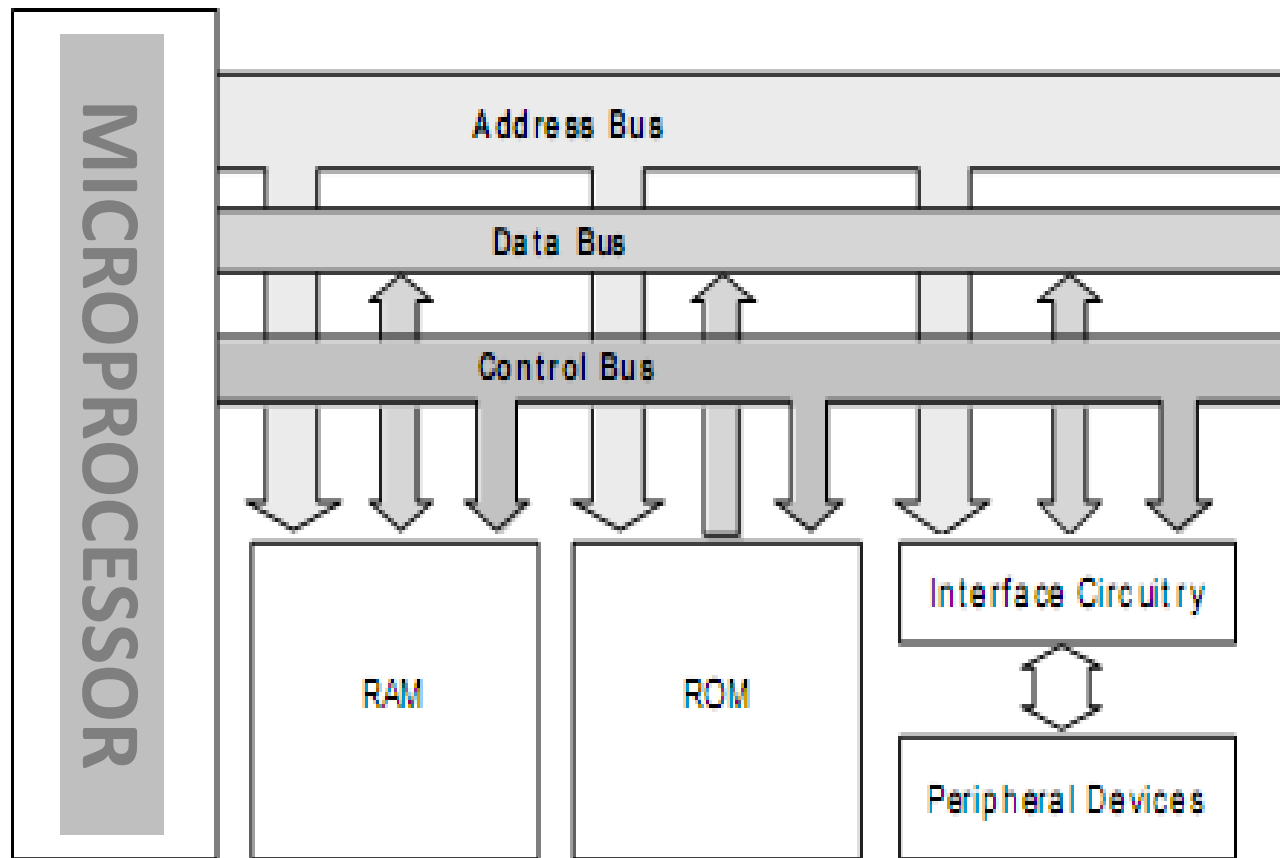
P1

- Microprocessor-based System
- Microcomputer: Detailed Block Diagram
- Introduction to Memory
- Memory Pin Connections
- SRAM Devices
- DRAM Devices
- ROM
- Memory Map
- Memory Address Decoding

Microprocessor-Based System



Microcomputer: Detailed Block Diagram



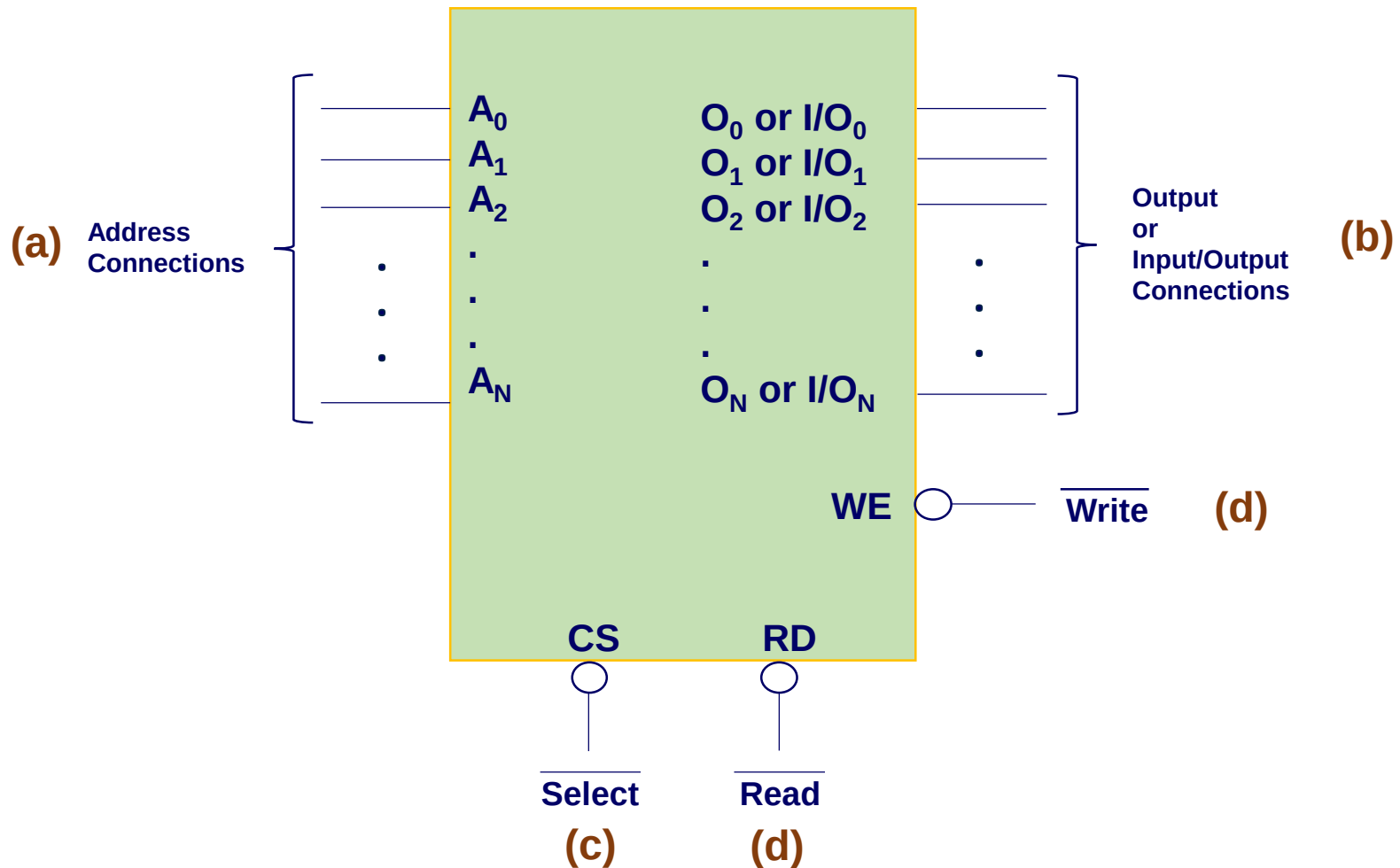
Introduction

- Whether **simple or complex**, every **microprocessor-based system** has a **memory system** including the Intel family of microprocessors.
- Almost all systems contain *two main types of memory*:
 - **Read-only memory (ROM)**
 - **Random-Access memory (RAM) or Read/Write memory.**
- **Read-Only Memory** contains **system software and permanent system data.**
- **Random-Access Memory** contains **temporary data and application software.**

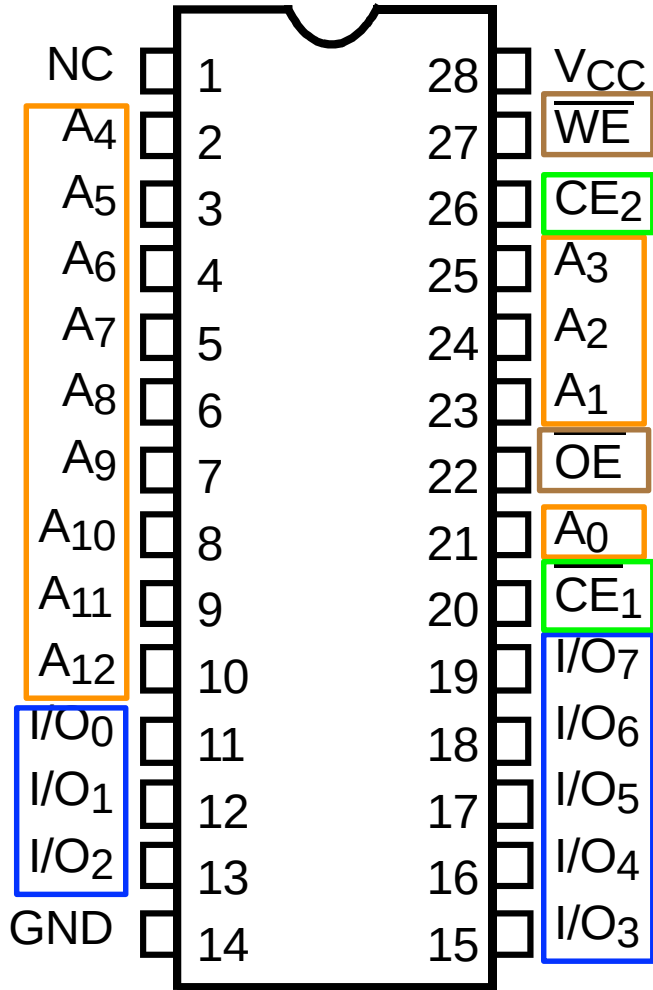
Memory Pin Connections

- Pin connections common to all memory devices are:
 - a. Address inputs*
 - b. Data outputs or inputs/outputs*
 - c. Some type of selection input*
 - d. At least one (1) control input used to select a read or a write operation*

Memory Pin Connections



Memory Pin Connections



a) address bus

b) data bus

c) chip select/enable

d) read/write enable

6264 (8K x 8) Static RAM)

Address Connections

- All memory devices have address inputs that select a memory location within the memory device.
- Address inputs are almost always labeled from A_0 , the *least significant address input*, to A_N where subscript **N** can be any value *but is always labeled as one less than the total number of address pins*.

Example 1

- **Given:**

A memory device has 11 address pins.

- **Required Tasks:**

- a) Label the 11 address pins of the given memory.
- b) Determine the number of memory locations in the given memory.

Example 2

- **Given:**

A memory device has 8192 memory locations.

- **Required Task:**

Determine the number of address pins in the given memory device.

Example 3

■ Given:

Consider the capacity of the following memory devices:

- 16K x 8
- 64K x 8
- 512K x 8

■ Required Task:

Determine the address bus width of each of the given memory devices.

Data Connections

- **All memory devices** have a **set of data outputs or input/output connections**.
- Many **memory devices** have **bidirectional common I/O pins** nowadays.
- The **data connections** are the points at which data are entered for storage or extracted for reading.
- **Data pins** are **labeled D0 through D7** for an **8-bit-wide memory device** and others.
- **Catalog listings of memory devices** often refer to **memory locations times bits per location**.

Selection Connections

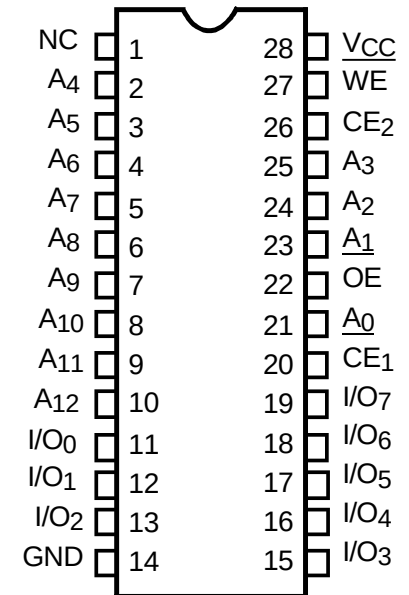
- Each memory device has an **input** (*sometimes more than one*) that **selects** or **enables** the memory device.
- This type of input is most often called a **chip select** ($\overline{\text{CS}}$), **chip enable** ($\overline{\text{CE}}$), or simply **select** ($\overline{\text{S}}$) input.
- **RAM** memory, generally, has at least one $\overline{\text{CS}}$ or $\overline{\text{S}}$ input.
- **ROM** memory has at least one $\overline{\text{CE}}$.

Control Connections

- All memory devices have some form of **control input** or inputs.
- **ROM** usually has only one control input often called the **output enable ($\overline{OE}$)** or **$\overline{R}$** , which *allows data to flow out of the output data pins* of the ROM.
- **RAM** memory device has either one or two control inputs;
 - if there is only one control input, it is often called **$R/\overline{W}$**
 - if there are two control inputs, they are usually labeled **$\overline{WE}$** (or **$\overline{W}$**), and **$\overline{OE}$** (or **$\overline{R}$**).

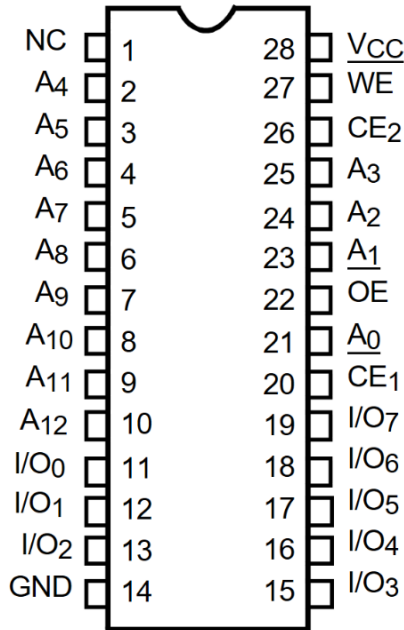
Static RAM (SRAM) Devices

- They **retain data** for as long as DC power is applied.
- They are also called **volatile memory** because they **will not retain data without power**.
- The main difference between RAM and ROM is that a **RAM** *is written under normal operation*, whereas a **ROM** *is programmed outside the computer and normally is only read*.
- **SRAM** are *usually constructed using transistors as memory cells*.



6264 (8K x 8) Static RAM

Static RAM (SRAM) Devices



6264 (8K x 8) Static RAM

PIN DESCRIPTIONS

Name	Type	Function
A0 – A12	Input	Address inputs for selecting one of the 8,192 x 8 bit words in the RAM
/CE1, CE2	Input	/CE1 is active LOW and CE2 is active HIGH. Both chip enables must be active when data read from or write to the device. If either chip enable is not active, the device is deselected and in a standby power down mode. The DQ pins will be in high impedance state when the device is deselected.
/WE	Input	The Write enable input is active LOW. It controls read and write operations. With the chip selected, when /WE is HIGH and /OE is LOW, output data will be present on the DQ pins, when /WE is LOW, the data present on the DQ pins will be written into the selected memory location.
/OE	Input	The output enable input is active LOW. If the output enable is active while the chip is selected and the write enable is inactive, data will be present on the DQ pins and they will be enabled. The DQ pins will be in the high impedance state when /OE is inactive.
DQ0~DQ7	I/O	These 8 bi-directional ports are used to read data from or write data into the RAM.
Vcc	Power	Power Supply
Gnd	Power	Ground
NC		No connection

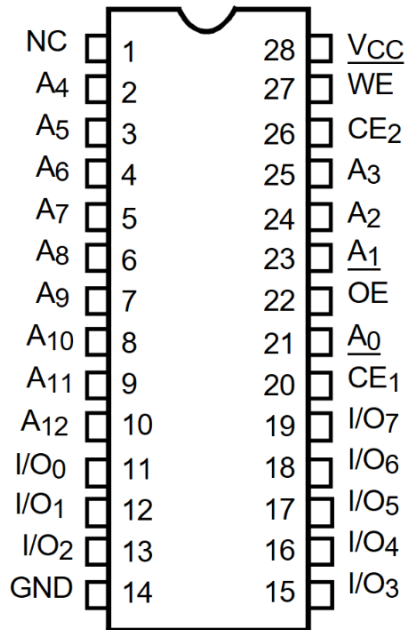
Datasheets:

<https://cdn-reichelt.de/documents/datenblatt/A300/HM6264.pdf>

<https://docs.rs-online.com/3630/0900766b8002aabd.pdf>

<https://ece-classes.usc.edu/ee459/library/datasheets/6264.pdf>

Static RAM (SRAM) Devices

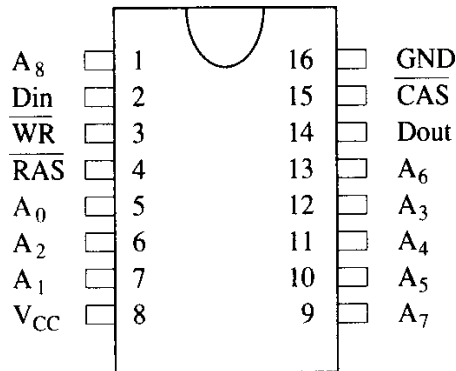


TRUTH TABLE

MODE	/CE1	CE2	/WE	/OE	DQ0~7	Vcc Current
Standby	H	X	X	X	High Z	I_{CCSB}, I_{CCSB1}
	X	L	X	X		
Output Disable	L	H	H	H	High Z	I_{CC}
Read	L	H	H	L	D_{OUT}	I_{CC}
Write	L	H	L	X	D_{IN}	I_{CC}

6264 (8K x 8) Static RAM

Dynamic RAM (DRAM) Devices



41256 - 256K x 1 Dynamic RAM

- They are essentially the same as SRAM, except that they retain data for only 2 or 4 ms on an integrated capacitor where it must be completely rewritten (refreshed) after.
- DRAM has far larger capacity than SRAM such as 256M x 8 or 2G bits *due to its construction using capacitors compared to transistors.*
- DRAM is used in PC memory modules.

Datasheet:

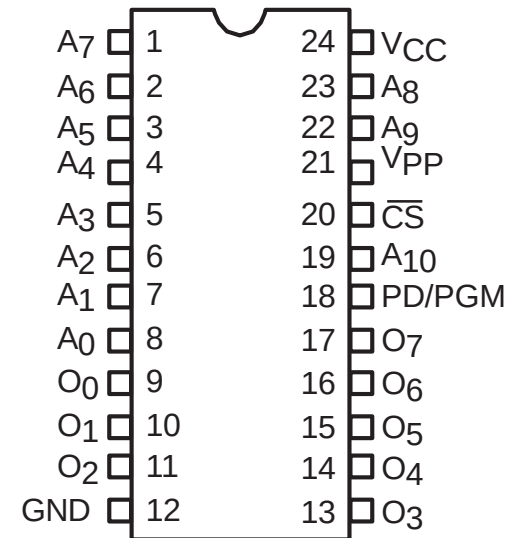
[https://colorcomputerarchive.com/repo/Documents/Datasheets/PD41256%20256k%20x%201bit%20DRAM%20PLCC%20\(NEC\).pdf](https://colorcomputerarchive.com/repo/Documents/Datasheets/PD41256%20256k%20x%201bit%20DRAM%20PLCC%20(NEC).pdf)

ROM

- The **Read-Only Memory (ROM)** permanently stores programs and data that are resident to the system and must not change when power supply is disconnected.
- The **ROM** is **permanently programmed** *so that **data** are always present, even when power is disconnected.*
- This type of memory is often called ***non-volatile memory***, because *its contents do not change even if power is disconnected.*

Types of ROM Devices: EPROM

- **Erasable Programmable Read-Only Memory**
- **Data can be erased (*via UV light*) and re-programmed (*via EPROM programmer*)**



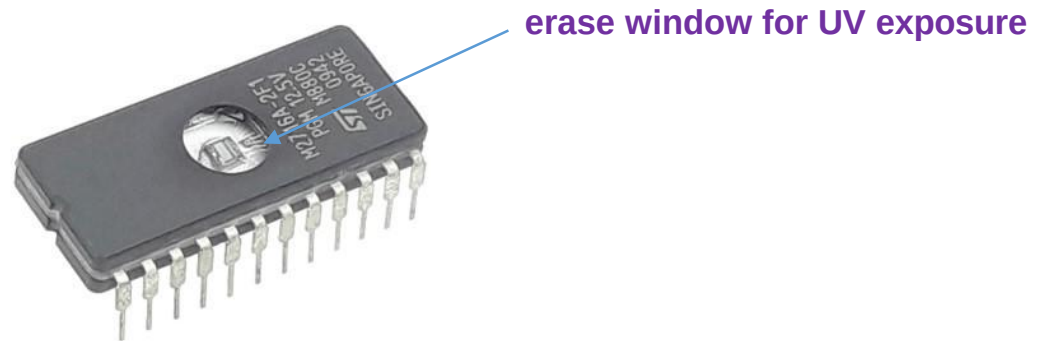
2716 (2K x 8) EPROM

Datasheets:

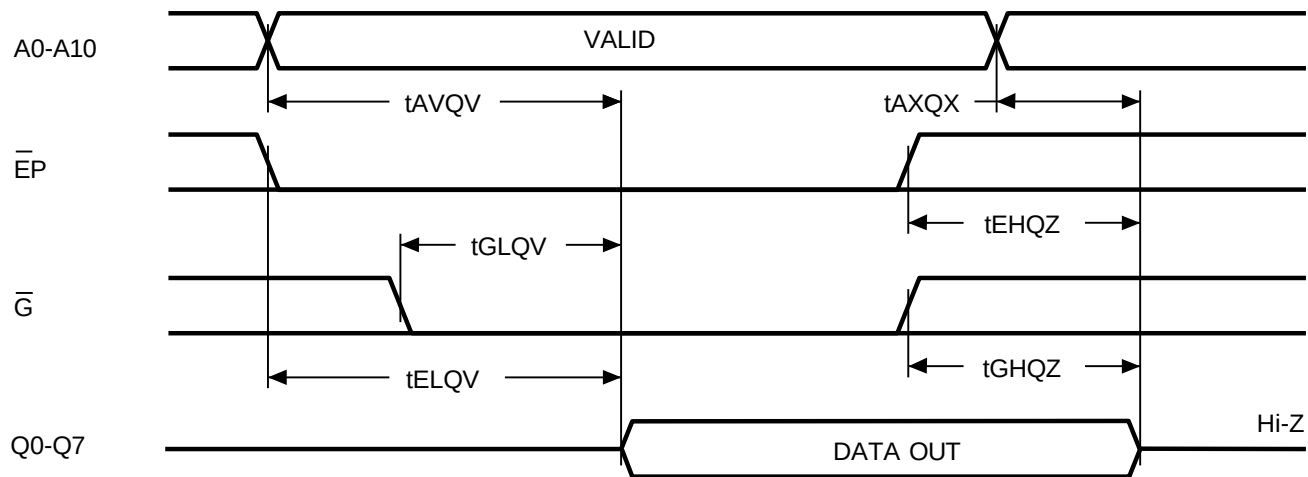
https://www.futurlec.com/Memory/2716_Datasheet.shtml

<https://downloads.reactivemicro.com/Electronics/ROM/2716%20EPROM%20-%201.pdf>

Types of ROM Devices: EPROM



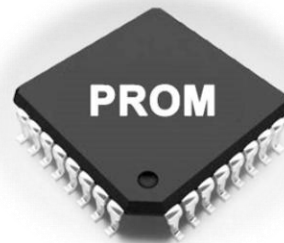
2716 (2K x 8) EPROM



2716 EPROM Read Cycle Timing Diagram

Types of ROM Devices: PROM

- **Programmable Read-Only Memory**
- **One-time Programmable ROM (OTP)**
- **Can be written to or programmed**
- **Once programmed, *data cannot be erased***



Datasheets:

https://en.wikipedia.org/wiki/Programmable_ROM

<https://www.ariat-tech.com/blog/what-is-prom-programmable-read-only-memory.html?srsId=AfmBOortl7ILq46wveWQ03WDyD9OQUZuRPV1ejVfZ-AoKIm81-rraaa>

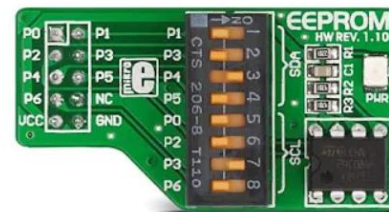
Types of ROM Devices: RMM

- **Read-Mostly Memory**
- It is also known as **flash memory** which includes the following:
 - ✓ **EEPROM**
 - ✓ **EAROM**
 - ✓ **NVRAM**

Types of ROM Devices: RMM

EEPROM

- **E**lectrically **E**rasable **P**rogrammable **R**ead-**O**nly **M**emory
- A non-volatile memory that retains data even when power is off
- Can be electrically erased and reprogrammed byte by byte
- Used to store small amounts of critical data, like device settings and calibration data, in electronics because of its ability to be updated without removing it from the circuit



Types of ROM Devices: RMM

EAROM

- Electrically **A**lterable **R**ead-**O**nly **M**emory
- A non-volatile memory that can be rewritten electrically but is much slower at writing than reading
- A type of EEPROM used for applications where data needs to be stored permanently but can be updated infrequently, such as storing critical system settings
- Unlike standard ROM, EAROM allows for the alteration of data, but its limited number of write cycles makes it best suited for infrequent rewrites.



Types of ROM Devices: RMM

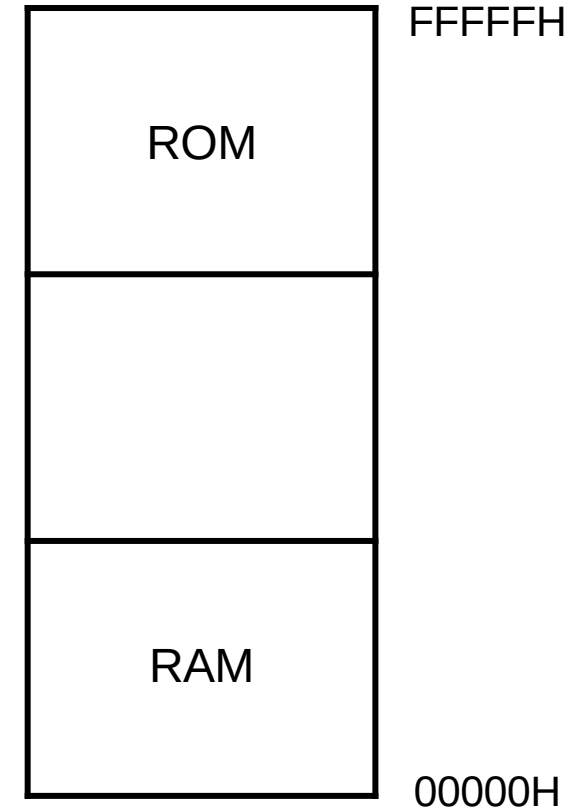
NVRAM

- **Non-Volatile Random-Access Memory**
- A type of memory that keeps its stored data even when the power supply is off, unlike standard volatile RAM.
- Combines the fast access of random-access memory with the persistent nature of non-volatile storage, making it ideal for storing critical system data like boot-up configurations, device settings, and passwords that need to survive power cycles



Memory Map

- The 8086 maps the RAM and ROM on the same address space.
- When **fetching an instruction**, the microprocessor has to **access the ROM**, or when **executing a push instruction**, it needs to **access the RAM**.



Memory Map for $2^{20} = 1\text{M}$

Memory Address Decoding

- Decoding makes the memory function at a unique section or partition of the memory map.
- Without an address decoder, only one memory device (either RAM or ROM) can be connected to a microprocessor, which would make it virtually useless.
- The 8086 microprocessor can address up to 2^{20} or 1M addressable space or total memory since it has 20 address lines ($A_0 - A_{19}$).
- Address range is from 00000H – FFFFFH.

Memory Map of 8086 μ P

ROM size* = 128K x 16

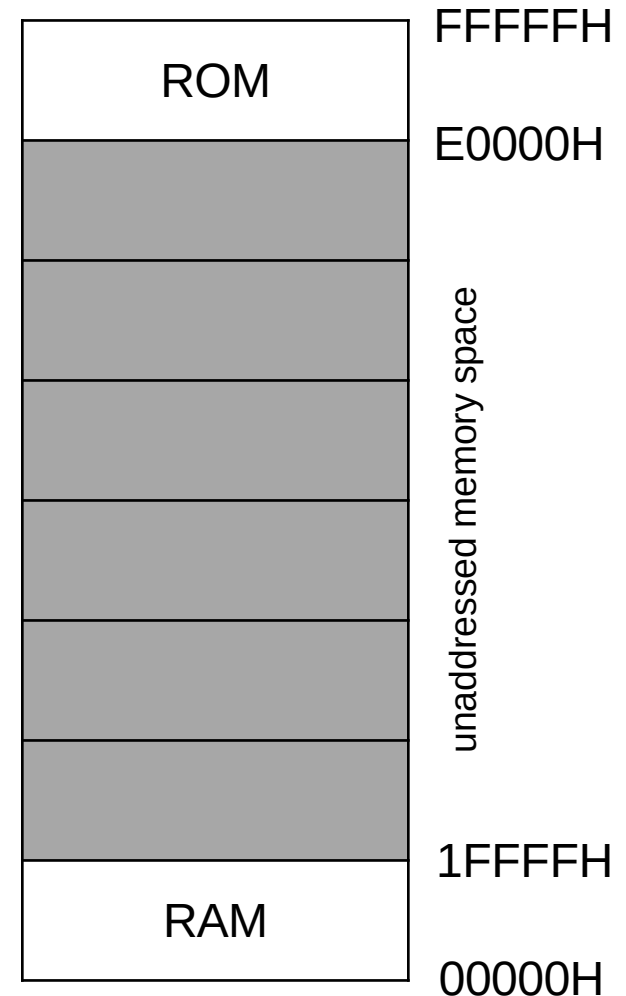
$$\begin{array}{cccccc} 0001 & 1111 & 1111 & 1111 & 1111 & = (128K - 1) \\ 1 & F & F & F & F & \end{array}$$

$$\begin{aligned} \text{Start Address} &= \text{FFFFFFH} - \text{1FFFFH} \\ &= \text{E0000H} \end{aligned}$$

RAM size* = 128K x 16

$$\begin{array}{cccccc} 0001 & 1111 & 1111 & 1111 & 1111 & = (128K - 1) \\ 1 & F & F & F & F & \end{array}$$

$$\begin{aligned} \text{End Address} &= \text{00000H} + \text{1FFFFH} \\ &= \text{1FFFFH} \end{aligned}$$



Memory Address Decoding

- The *objective of memory address decoding* is to enable the memory chips (RAM and ROM) when the address belong to either address range.
- For example, the address 001FFH will enable only the RAM chips since it is an address between 00000H – 1FFFFH. Address E00C2H will enable only the ROM chip.
- However, address A0000H does not enable either RAM or ROM since it belongs to the unaddressed portion in the memory.

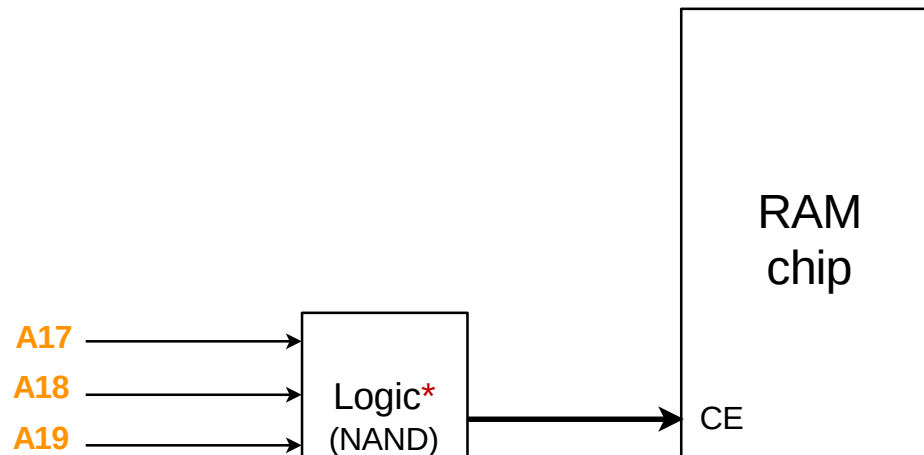
Memory Address Decoding Circuitry

- A decoding circuitry is needed for both RAM and ROM since they have different address ranges.
- The circuit will decode the address and determines if the address belongs to the address range of the RAM or the ROM.
- Only one of either RAM or ROM will be enabled depending on the address.

Decoding RAM Address

	19	18	17				0	
Start Address:	0000	0000	0000	0000	0000	0000		(00000H)
End Address:	0001	1111	1111	1111	1111	1111		(1FFFFH)
	000X	XXXX	XXXX	XXXX	XXXX	XXXX		

A17, A18 & A19 should be logic-0 for the address to be valid

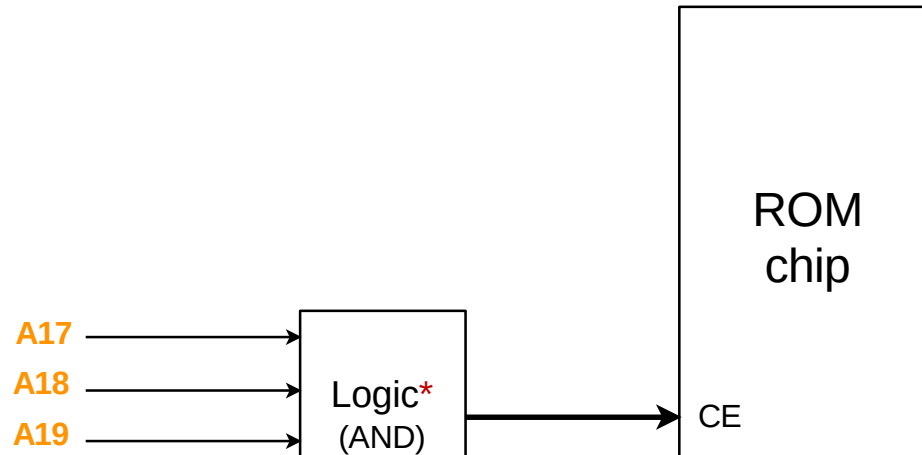


* The decode logic depends on the chip enable signal of the memory IC.

Decoding ROM Address

Start Address: ^{19 18 17} 1110 0000 0000 0000 0000 ⁰ (E0000H)
End Address: 1111 1111 1111 1111 1111 (FFFFFFH)
111X XXXX XXXX XXXX XXXX

A17, A18 & A19 should be logic-1 for the address to be valid



* The decode logic depends on the chip enable signal of the memory IC.



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CpE 3104 – Microprocessors

End of Lecture

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